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ABILITY ENHANCEMENT COURSE
REPORT ON LATEX
FOR

"WEBSITE CONTENT SUMMARIZER TO BRAILLE CONVERSION"

Submitted in partial fulfillment of the requirements for the award of

degree of Bachelor of Engineering in

COMPUTER SCIENCE & ENGINEERING

Submitted by

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Under the Guidance of

Mrs. Sophia G
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MVJ COLLEGE OF ENGINEERING
BANGALORE-67

ACADEMIC YEAR 2024-25

MVJ COLLEGE OF ENGINEERING

Near ITPB, Whitefield, Bangalore-67

(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Ability Enhancement Course project work titled **'WEBSITE CONTENT SUMMARIZER TO BRAILLE CONVERSION'** is carried out by **KARTHIK SHETTY K (1MJ22CS408), NISAR AHAMED (1MJ22CS410), RAJDURAI D KUDIYANNAVAR (1MJ22CS414)** and **WILLIAMS KA (1MJ22CS423)** who are Bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfilment for the award of Degree of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2024- 2025. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the Ability Enhancement Course project report deposited in the departmental library. The Ability Enhancement Course project report has been approved as it satisfies the academic requirements in respect of Ability Enhancement Course project work prescribed by the institution for the said Degree.

Signature of Guide	**Signature of HOD**	**Signature of Principal**
Mrs. Sophia G	**Dr. Kiran Babu T. S**	**Dr. Ajayan K R**
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DECLARATION

We, **KARTHIK SHETTY K (1MJ22CS408), NISAR AHAMED (1MJ22CS410), RAJDURAI D KUDIYANNAVAR (1MJ22CS414)** and **WILLIAMS KA (1MJ22CS423)** students of Eighth semester B.E., Department of Computer science and Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the Ability Enhancement Course project titled **'WEBSITE CONTENT SUMMARIZER TO BRAILLE CONVERSION'** has been carried out by us and submitted in partial fulfilment for the award of Degree of **Bachelor of Engineering** in **Computer Science and Engineering** during the year 2024-2025. Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

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ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany a successful completion of any task would be incomplete without the mention of people who made it possible, success is the epitome of hard work and perseverance, but steadfast of all is encouraging guidance.

So, with gratitude we acknowledge all those whose guidance and encouragement served as beacon of light and crowned our effort with success.

We are thankful to **Dr. Ajayan K R, Principal of MVJCE** for his encouragement and support throughout the Ability Enhancement Course project work.

We are thankful to **Dr. Asha Joseph, Dean, School of CSE, MVJCE** for her encouragement and support throughout the Ability Enhancement Course project work.

We are thankful to **Mr. Kumar**, **Controller Of Examinations** for his encouragement and support throughout the Ability Enhancement Course project work.

We are also thankful to **Dr. Kiran Babu T S, HOD, CSE Department** for his incessant encouragement & all the help during the Ability Enhancement Course project work.

We consider it a privilege and honor to express our sincere gratitude to our guide **Mrs. Sophia G, Assistant Professor, CSE Department** for her valuable guidance throughout the tenure of this Ability Enhancement Course project work, and whose support and encouragement made this work possible.

It is also an immense pleasure to express our deepest gratitude to all faculty members of our department for their cooperation and constructive criticism offered, which helped us a lot during our Ability Enhancement Course project work.

Finally, we would like to thank all our family members and friends whose encouragement and support was invaluable.

Thanking You

Website Content Summarizer To Braille Conversion

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***Abstract*—The rapid expansion of digital content has made information widely accessible, but visually impaired individuals often face significant challenges in navigating and understanding online information. While screen readers provide basic accessibility, they lack efficiency in summarizing content and are not designed to cater to Braille users effectively.This project proposes a "Website Content Summarizer to Braille Conversion" system that bridges this gap by automating the summarization of website content and converting the extracted summary into Braille. The system leverages Natural Language Processing (NLP) techniques for summarization and Braille encoding algorithms for conversion. By delivering concise and relevant information in Braille, this solution aims to enhance accessibility and ensure inclusivity for visually impaired individuals.The proposed system focuses on user-friendly functionality, support for multiple languages, and adaptability to various content types, making it a valuable tool for enabling equal access to digital information.**

***Index Terms*—Braille Conversion,Text Summarization,Web Accessibility,Multilingual Translation,Text-to-Speech (TTS).**

I. Introduction

In Braille writing system, dots are embossed on paper. A group of dots represent certain character of any specific language, which is perceivable by the finger tip. Optical Braille Recognition process takes image of these tactile alphabets, and translates them to plain text. Visually impaired people have all the abilities that any normal person need, to contribute to the society. The only problem they face is not being able to visually perceive the world. As most of our learning procedures are largely dependent on the eye, a big portion of this human potential is not getting the room for manifestation. So rather becoming an asset, they turn into a burden to the community. This problem was intensely severe till the early 1800's, when Braille Writing system came into existence. The base process was developed by the French Army for passing message in the dark. The system was then matured by a young boy named Braille, who accidentally lost his vision at an early age. He simplified the writing and reading process, and as a result the technique became very popular among visually impaired people. Further refinement to the procedure shaped it for mass use. Now, almost every major language in this world has its own Braille format. Braille is a specialized writing system for visually impaired people. Here raised dots on embossed paper are used as tactile alphabet [1]. Visually impaired people are an integral part of the society. However, their disabilities have made them to have less access to computers and Internet than the people with clear vision. Over time Braille system has been used by them for written communication. Braille is a system of writing that uses patterns of raised dots to inscribe characters on paper. This allows visually impaired people to read and write using touch instead of vision. It is the way for blind people to participate in a literate culture. First developed in the nineteenth century, Braille has become the pre-eminent tactile alphabet. Its characters are six- dot cells, two wide by three tall. Each dot may exist or may not exist giving two possibilities for each dot cell. Any of the six dots may or may not be raised; giving 64 possible characters. In English it includes 26 English alphabets, punctuations, numbers etc [2].Languages are the most historical manner of conversation between people, either thru speech or written form. Recently, visible textual content in natural or synthetic scenes can convey very important and beneficial information. So scientists began to digitize these pictures, extract and interpret the records the usage of special techniques, and then whole textual content-to- speech (TTS). This is performed to read the statistics for the ease and comfort of the user. Text extraction and TTS can be used together to assist humans with reading and hearing impairments read written data using a pc gadget . In this, paintings, a new textual content detection framework based on joint factor analysis is proposed, and the MSER set of rules is used to extract which are universal as candidate letters. Which may be characters, are decided on consistent with their geometric houses and versions in stroke

width. Next, the selected items are grouped into the following detected texts, which are then broken down into man or woman words. Optical person reputation is used to understand and extract words, and eventually the extracted textual content is converted into suitable speech the usage of a synthesizer. The proposed set of rules is examined on photographs representing diverse scenes, from documents to nature scenes[3]. Many valuable documents are scanned and kept for backup. Converting the scanned text images to machine readable text is termed as Optical Character Recognition (OCR) and this topic have always been an interesting topic in Computer Science. Braille has been the primary reading and writing system for the Visually Impaired since 19th century. A tool is always required that can be used to extract text from the images using the powerful Open-Source Tesseract OCR Engine and then apply it in Braille Translation by converting the extracted text to Braille. The application is thus useful for the visually impaired as it can make Braille texts from scanned image documents and can be used for translating old valuable books to Braille format[4]. The raised dot system now known as Braille. It was established by a Frenchman Louis Braille in 1824.louise had found 63 ways to used six dot cells in an area no larger than a fingertip. It took 2 years after his death to adopt as official communication language for blind person .Today braille is used throughout the world and is taught the children from early age. Braille is used for reading and writing. The person who read braille will fill the dots under their fingertips. Braille letters are made up of six raised dots arrange in group called a cells. Like one half of domino. Each letter of the alphabets is made up of different combination of these six dots. Which make up word. Blindness is considered to be highest among all other disabilities. More than 15 million people in India are blind. Blind people faced tremendous difficulties in accessing information from document. Thus in order to build communication gap between blind person and community's braille system is used [5]. Braille is writing system invented by Louis Braille which uses the sense of touch. The reader caresses over the dots with its fingertips to recognize the pattern of dots. Every character, symbols, word of any other language are mapped to unique pattern dot. These dots are raised on a paper. A standard braille page is 11 x 11 inches with a maximum of 25 lines per page accommodating 40-42 braille cells per line. The dot height is 0.5 mm and horizontal and vertical spacing is about 2.5 mm between the dots. The whitespace between the dots of the adjacent cell horizontally is 3.5 mm and 5 mm vertically. There are two formats for braille: single side of a sheet or both side of a braille sheet (inter-point braille). More information can be accommodated on the inter-point braille, where the dots are punched on both sides of the sheet with a slight diagonal offset. According to the encoding used the braille can be divided into three categories. The grade 1 braille, corresponds each cell to a character. Grade 2 braille aims to represent each cell to a group of characters. Grade 3 braille uses their own dialect of code. It is a non-standardized format. This project aims to convert regional Hindi language braille to its speech. This project comes under the domain of natural language processing. Optical braille recognition system involves Braille image acquisition, braille image pre-processing, braille dot segmentation, braille dot recognition and conversion into natural language characters [6].India has a lion share of blind people of the world and it also has one of the highest percentages of unemployable and illiterate blind people. Therefore Bharati Braille is a language of utmost importance for the upliftment of blind people in India. But unfortunately, there is a scarcity of resources like books, schools and even teachers to teach blind people. It is very time cumbersome and complex task to write books in Braille and even costly. However with the recent advancement in the field of technology it is possible to reduce the complexity and cost of teaching blind people. The automatic conversion of Braille documents into natural languages and speech helps any arbitrary tutor to teach blind students without even knowing Braille fluently. This reduces the need of special tutors to teach Braille. The automatic conversion of documents in natural languages to Braille helps in printing books written in any natural language into Braille. It is also possible to teach the blind students using a refreshable Braille slate which can refresh the dot pattern after taking input directly from the optical character recognizer and read the text like any non-blind person would do. This conversion of Braille into natural language and vice versa is called as Braille transliteration. There are many works done in the field of Braille transliteration of languages like English, Arabic, etc. However there is scarcity of work in the field of Braille transliteration of Indian local languages like Odia. The work presented in [1] describes a realistic method of Braille transliteration of printable text in Odia. The transliterator reads the letters from the text file and converts each letter into Braille using Unicode mapping to Braille code. The Braille codes are mapped to corresponding Braille cells and then the Braille cells are organized into a standard Braille sheet. However there is still no method which can be used to make the conversion of printed sheets into Odia Braille or handwritten sheets into Odia Braille. This task has become physically realizable with considerably good accuracy because of the design of high accuracy Optical character recognizer (OCR) for Odia language. The present paper proposes a method to automatically convert handwritten characters into Braille, which can then further be used to convert handwritten texts and printed texts into Braille. The method proposes a way to convert Odia handwritten texts into Braille as follows: take an image of the text as input and convert it into a printable text format by using Odia OCR, convert characters in the printable text into the corresponding Unicode and the map the Unicode to corresponding Braille code and finally to Braille cell.Braille consists of cells of six dots arranged in two columns of three dots each. Different combinations of dots represent different letters, numbers, and symbols. Braille is used worldwide as a standard writing system for people who are blind or visually impaired, and it is widely used for reading and writing in a variety of languages. The Braille system allows people who are blind or visually impaired to participate in activities such as reading, writing, and studying,

which would otherwise be difficult or impossible for them, and it helps to break down barriers to information and education that are faced by this community. Bharti Braille is a form of Braille script used in India to write the languages of India in Braille script. It is based on the standard Braille script but includes additional characters to represent the unique sounds and characters found in Indian languages. The literary script is based on the standard Braille script but includes additional characters to represent the unique sounds and characters found in Indian languages. This includes characters for the retroflex sounds found in many Indian languages, as well as special characters to represent the unique conjunct consonants found in Indian languages [8]. Visually impaired people are an important part of every community. They are also concerned in learning the details of everything they encounter in their daily life .The total number of visually impaired people is 2.2 billion; among them, 36 million are completely blind, and rest of the 1 billion have moderate to severe vision impairment. Approximately thirty-seven million of the six billion populations worldwide are suffering from blindness. Accessibility issues have been an important research domain over the last few years promoting the development of thousands of smartphone applications to help people with a visual disability, e.g., voiceOver services, talkback services, screen readers, and navigators. These researches resulted in a dramatic increase in mobile-screen reader usage for the visually impaired[9]. Eye diseases, ageing, car accidents, and other causes contribute to an annual increase in the number of persons who are blind or visually impaired. Because reading is such an important part of human life (text can be found in newspapers, commercial products, signboards, computer screens, and so on), visually impaired people encounter numerous challenges. Visually impaired people can use mobile applications to assist them in reading text. The goal of our study is to see if a visually impaired person can acquire auditory information regarding comes to taking up a lot of space and using a lot of power. Threshold inverting quantization (TIQ) comparator can be printed text, text boards, scene text, hoardings, and traffic signboard instructions. To attain the best results and reach state-of-the-art. Nearly half of these cases, or at least 1 billion, involve vision damage that either might have been prevented or is still unaddressed. The majority of those who are blind or have visual problems are over 50, however visual impairment can affect anyone at any age. Louis Braille created Braille in 1824 as a reading and writing system for the persons who are blind. The basic Braille system consists of six raised dots that can be combined to form 64 different signs. Blind or visually impaired people can read embossed characters by feeling them with their fingers. In the Braille language, each character has a distinct code that makes it easier to distinguish between them. An estimated 90 percent of the 37 million blind persons worldwide reside in developing nations. India is home to about 15 million of them[11]. India is the largest country where a large amount of visually impaired people can see. We have 2.2 billion people, among them, 36 million are blind and the rest of 1 billion people are having modest to secure vision impairment.Blindness can be permanent or temporary. If any portion of the eye gets damaged leads to blindness and also if any effects causes in the brain can also lead to blindness. By electronic gadgets like smart-phone, tablets, PC, laptops, where these gadgets will emit light and it has found negative effects on our eyes. Where it has the potential to lead to macular degeneration by passing to the pupil and cornea to beam directly into the retina.A per researcher, one-third of smartphones are using for educational purposes and their usage has been increasing day by day. And it has a very low presence of humans with viral disabilities Braille is the language that is used for visually impaired people. Even they can read and write as normal humans do. Where it includes wax tablets, manipulation of wooden letters, etc. The main disadvantage here is that each required material is extremely hard to make, and also, very infeasible for someone to hold and carry them. After the technology is developed, a new method starts introducing and also improving. Valentin Havy is the founder and inventor of the first blind school in the world, and by on heavy paper using metal pen sounded tip he produced raised letters. Whatever system he produced for raised letters underwent several modifications[12].The World Health Organization projected that around the globe, roughly 285 million individuals encounter varying degrees of visual impairment, with 39 million of them falling under the classification of blindness. A noteworthy segment of this population relies predominantly on Braille as their primary means of participating in reading and writing endeavors. The majority of Braille materials are available only in a limited number of languages, hindering access to the information for those who speak languages not widely represented in Braille format. Braille system, which relies on tactile reading and writing using the fingers, is utilized by the blind for communication and access to written material[13].Nowadays, information can be gained by various platforms and media it could be social media platforms, radio channels or new channels, etc. very huge amount of information is consumed by various users across the world. As nowadays a lot of content is consumed using YouTube. According to Buffer Library (blog), YouTube is the world's second most used social media site having 2.2 billion Monthly Active Users and Facebook is in first place having 2.8 billion Monthly Active Users. On YouTube, it is very easy to watch videos whether it is for educational purposes or entertainment purposes the user needs to find the relevant content of any genre the user wants to consume. After a successful search of the content on YouTube, the user wants to view it. A high number of users get a lot of problems while watching the content which such as network problem which restricts the user to watch the content in very low quality which leads the content to blurry and loading of the video might be sometimes frustrating also there are also a lot of clickbait videos created for earning purpose only, having lack of information in it. There are numerous situations where people click on the wrong videos because of their fancy thumbnail and titles wasting all their time gaining wrong or useless information. Students mostly these

days prefer to clear all the topics which they are unaware of and also want to prepare for exams in very less time for that they fast forward the videos and try to understand the concept in very less time because of this they usually get confused in topics which they were watching. Sometimes working professionals want to know what work or terms were discussed in the meeting which was highly time-consuming in very less time they can use the idea of using summarized video text in it, so working professionals can easily save their time very effectively. Talking about text summarization, first, a summary is a brief statement or restatement of the main point's important information of the text in a shorter form. The advantage of text summarization can be a decrease in time for the users to get the information precisely[14].With the steady advancements in the field of technology,the internet growth has also increased to a great extent. In today's fast moving world, information is in great abundance.Information is easily available in the form of text documents,files, statistics and data. As the internet contains more than the required amount of information, it becomes very difficult to search for valuable information through a vast quantity of documents. People need to take decisions and even need to acquire knowledge quickly and on the go. Nowadays, even 24 hours are not enough to complete our day to day tasks and having to read pages and pages of emails, books or even websites would be impossible. To help aid the situation, we have designed an application that will help generate a summary or a brief version of the text provided to the application. Our application mainly emphasizes taking transcripts or data from the user and giving a summary of the transcript or data. User can either enter his own text or enter the URL from where he wants the data to be summarized. On the basis of the data received, we provide an appropriate summary for the same.This will be very beneficial as it will help users to get relevant information from a large pool of data. The user need not read the entire document but just the summary generated by our application to know the content present in the data. It will help users to get a brief idea of the entire document based on the summary generated by our application. The summary provided by our application ensures to include the most relevant information such that the user does not miss on any important points. This will help save on time and gain maximum information from the data.[15]

II. Related Works

In order to improve accessibility for the blind and visually handicapped, this research proposes a revolutionary system for translating text from major Indian languages to Bharti Braille. To handle ambiguities in the conversion process, the suggested system uses a hybrid strategy that combines rule-based translation with an LSTM model. Additional characters are included in India's Bharti Braille tactile writing system to represent distinctive Indian language sounds and symbols. The rule-based part of the system creates mappings between the characters of the Indian language and their equivalents in Bharti Braille. However, simple transliteration frequently results in errors because of the intricate structure of scripts

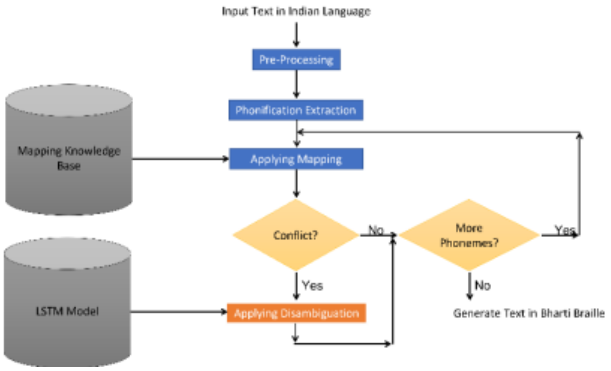


Fig. 1. Working Of Indian Languages To Bharati Braille Translation System

like Devanagari, which consists of vowels, consonants, and diacritical markings. The LSTM model is used to eliminate ambiguities and guarantee accurate Braille encoding in order to lessen this.After extensive testing in ten main Indian languages, the system's translation accuracy was close to 100 percent.By offering a precise and effective way to convert text to Braille, this device greatly enhances the visually impaired's ability to access information. It also emphasizes how crucial user training is to making the most of Braille conversion technologies. All things considered, this endeavor is a major step toward improving the accessibility of written material for India's blind and VI population.

III. Results and Discussions

The Website Content Summarizer to Braille Conversion program is a versatile and innovative tool designed to transform how users access and interact with online information. By effectively extracting and processing textual content from URLs, the program addresses the need for making digital content more accessible to diverse user groups, including visually impaired individuals. Its features encompass a multi-step workflow that begins with content extraction, where relevant text is retrieved from web pages and cleaned to remove unnecessary elements. The text is then summarized using a frequency-based algorithm that identifies key information, condensing lengthy content into concise, meaningful summaries without compromising context or accuracy. This summarized text is further processed for Braille conversion, employing language-specific mappings to produce outputs in Bharti Braille for Hindi, Kannada, and English.

In addition to its Braille capabilities, the program supports multilingual translation, allowing summaries to be converted into several languages using advanced translation tools. This feature ensures inclusivity by catering to users who prefer or require content in languages other than the original. Furthermore, the program incorporates a text-to-speech conversion module that generates audio files for auditory playback, enabling users to listen to summarized content. The audio files are stored and served through a user-friendly web interface, which streamlines the process of inputting URLs, selecting output languages, and accessing results. Outputs provided by

the program include the summarized text, Braille representation, translated summaries, and downloadable audio files, ensuring a comprehensive and accessible experience.

The program excels in accuracy and efficiency, with quick processing times that enhance user satisfaction. Robust error handling ensures that invalid URLs or unexpected inputs are managed gracefully, maintaining reliability across diverse use cases. The user interface is intuitive, minimizing the technical barriers for users and enabling easy navigation through its features. By combining content extraction, natural language processing, translation, Braille conversion, and audio generation, the program achieves its goal of delivering web content in multiple accessible formats, ultimately empowering users and contributing to a more inclusive digital landscape. The versatility and adaptability of the program make it a valuable tool in bridging the gap between technology and accessibility, with potential for further enhancements to expand its scope and impact.

IV. Conclusion

The Website Content Summarizer to Braille Conversion program is a groundbreaking initiative designed to enhance digital accessibility for visually impaired users. By combining web content extraction, text summarization, multilingual translation, Bharti Braille conversion, and text-to-speech capabilities, it provides a seamless end-to-end solution for transforming online information into accessible formats. The program efficiently extracts and condenses content from user-provided URLs, translates it into various languages, and converts it into Braille and audio outputs, catering to diverse user needs. This integration of natural language processing, machine learning, and assistive technologies ensures that complex and lengthy web content becomes concise, understandable, and accessible to all.

While the program addresses critical gaps in accessibility, there remains significant potential for growth. Future advancements could include employing deep learning techniques to enhance summarization accuracy, expanding support for more languages in Braille and audio formats, optimizing computational performance for real-time use, and incorporating robust mechanisms to detect and mitigate security risks from malicious content. This program highlights the power of interdisciplinary approaches in solving real-world problems and stands as a testament to how technology can create a more inclusive digital environment for users with diverse needs.

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